Worksheet 2-1 A
Average and Instantaneous Rates of Change #1
AP Calculus

Name_____

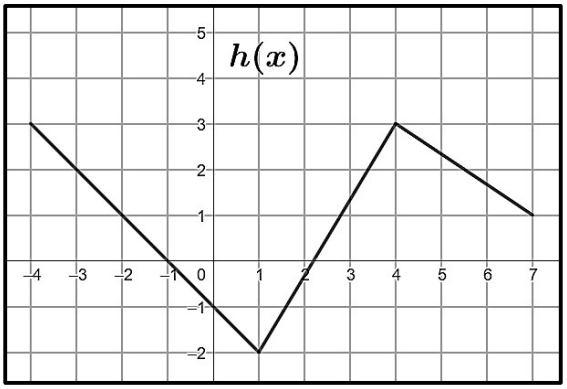
For questions 1 – 3: find the average rate of change of the functions on the interval [1, 4].

1. $g(x) = \frac{8}{x} - 3$

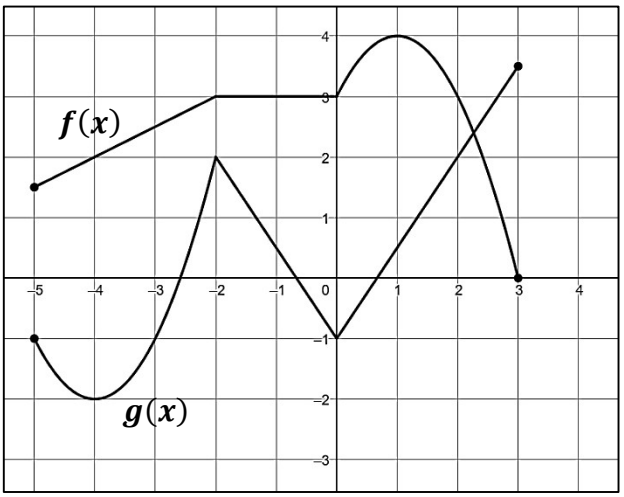
2.

x	0	1	3	4
$p(x)$	-1	-3	0	5

3.



The graphs of f and g are given below. For each function, find the average rate of change on the given interval.



4. $h(x) = f(x) + g(x)$ on $[-4, 3]$

5. $k(x) = f(g(x))$ on $[-4, 0]$

6. $w(x) = g(f(x))$ on $[-2, 3]$

7. Mr. Carlson leaves his house at 6:35AM to head to work. He arrives at school at 7:03AM. The trip from his house to school is 19 miles. Find the average velocity, in miles per hour, for his trip.

Find the instantaneous rate of change of each function at the given x-value.

8. $f(x) = \frac{1}{3x}$ at $x = 4$

9. $g(x) = 2x^2 + 1$ at $x = -2$

10. $h(x) = 7\sqrt{x}$ at $x = 2$

11. $p(x) = 5x - 2x^2$ at $x = -2$

Each limit represents the instantaneous rate of change of a function. Identify the original function and the x–value of the instantaneous rate of change.

12. $\lim_{x \to 1} \frac{\frac{1}{10+x} - \frac{1}{11}}{x-1}$

Function: $f(x) =$

Instantaneous Rate at $x =$ ____

13. $\lim_{h \to 0} \frac{2(6+h)^2 + (6+h) - 3 - (75)}{h}$

Function: $f(x) =$

Instantaneous Rate at $x =$ ____